

Swiss Institute of Bioinformatics

SINGLE-CELL TRANSCRIPTOMICS WITH R

Introduction

Bellinzona

Nov 11-13, 2026

elixir SWITZERLAND

All SIB courses abide the [ELIXIR Code of Conduct]

We **value** each other's perspectives providing a safe environment for people to be themselves.

We will **maintain** high ethical standards across all ELIXIR events.

We **adopt** a zero-tolerance approach to harassment and discrimination in any form.

We will **apply** honesty and integrity in the dealing of any transgressions against the Code.

We are **committed** to making ELIXIR events a collaborative, supportive and enjoyable experience.

We will **ensure** that our environment allows everyone to feel respected and included.

https://elixir-europe.org/events/code-of-conduct

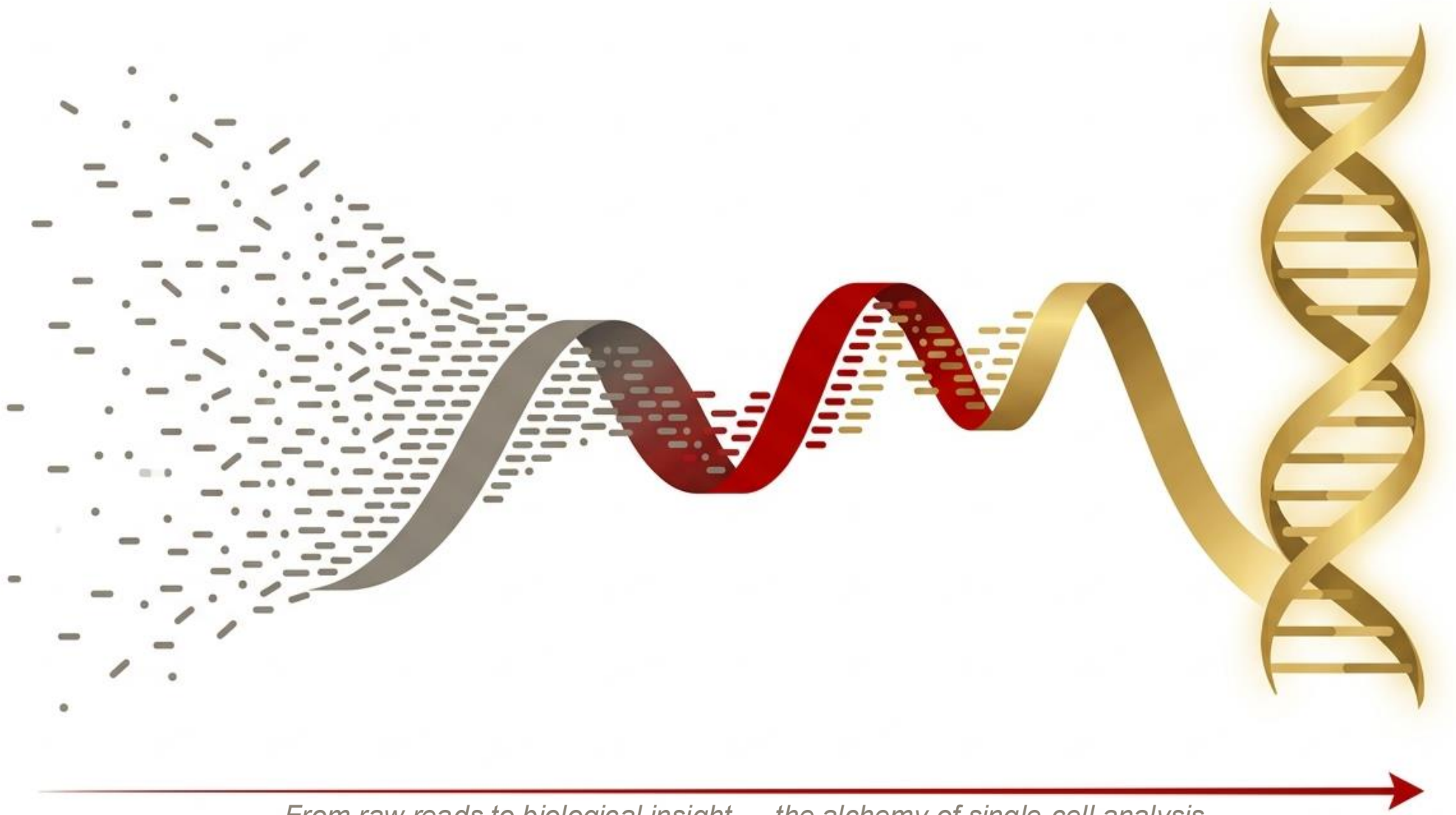
Learning outcomes

- Explain the characteristics of the most-used methods to scRNA-seq data
- Perform quality control at different steps of the analysis
- Use dimensionality reduction (PCA, t-SNE, UMAP)
- Apply best practices for translating single cell transcriptomic data into biological knowledge

Learning experiences

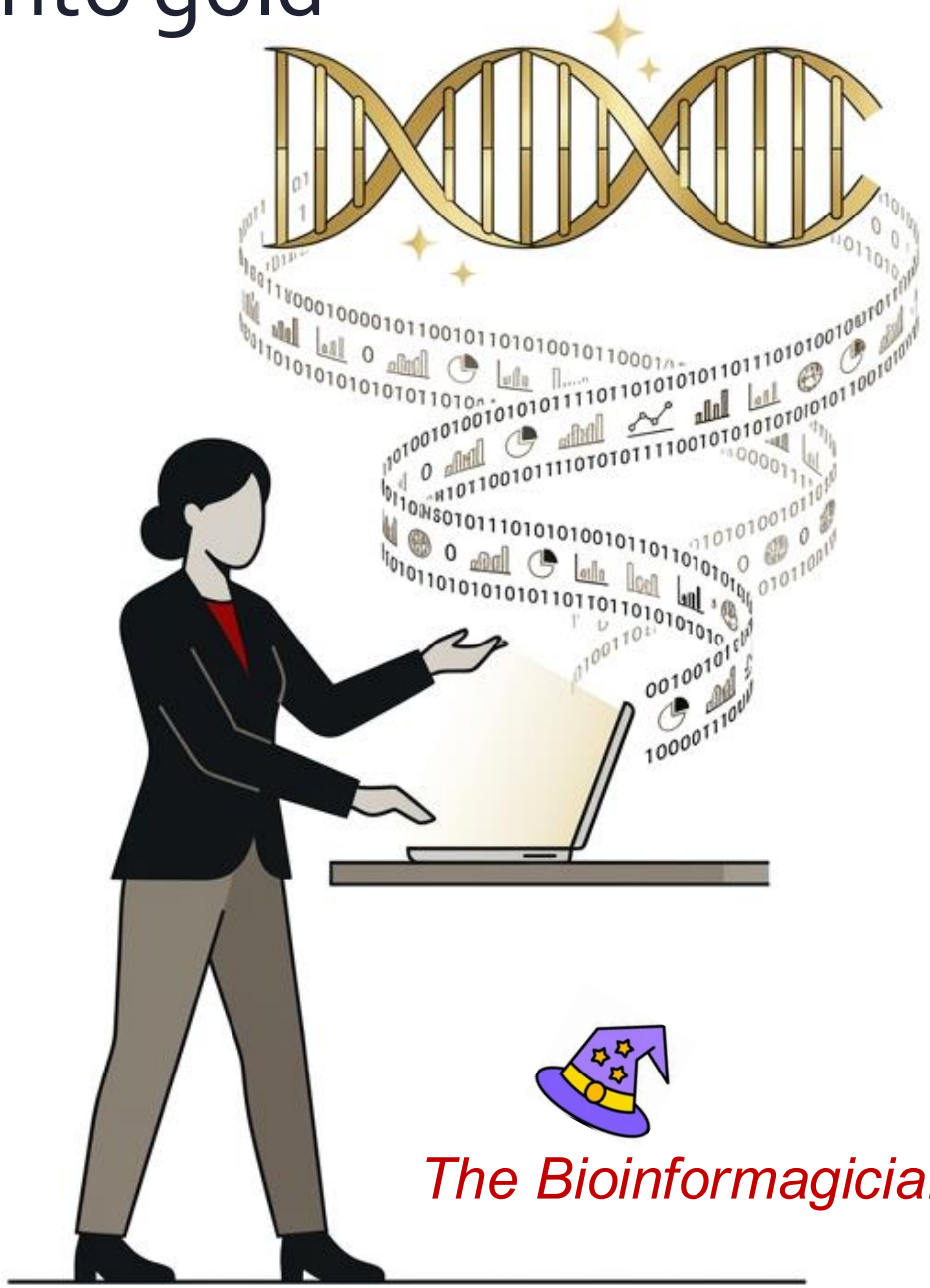
- Lectures
- Quizzes
- Exercises
- Group projects
- Exam

Transform data into gold



From raw reads to biological insight — the alchemy of single-cell analysis

Transform data into gold

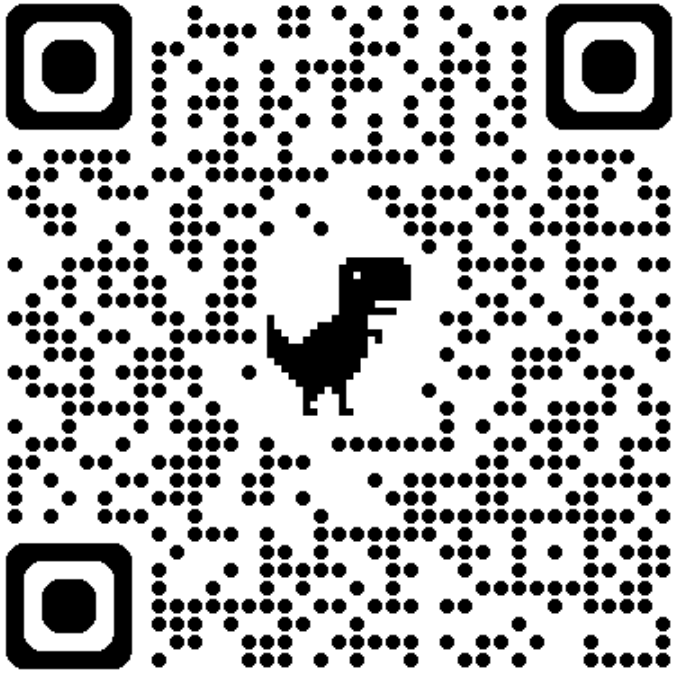


The Bioinformagician at work

Communication

- Course website: https://sib-swiss.github.io/single-cell-r-training

- Shared document
  - General links for the course
  - Please write your name for the exam



About me – Luciano Cascione



Background in computational science and bioinformatics



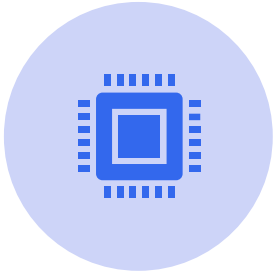
Co-Head of the Bioinformatics Core, Institute of Oncology Research (IOR)



Research focus: B-cell lymphomas, RNA-Seq & single-cell omics



Passionate about teaching, reproducible and impactful data analysis



Developer of bioinformatics tools (e.g., EasyCircR for circRNA analysis)



Bike-aholic, married, two kids

The Bioinformatics Core Unit



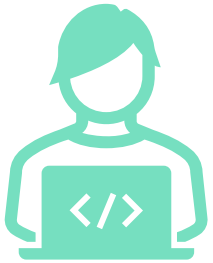
Making advanced computational analysis accessible to

9 IOR-Groups
+
IELSG, EOC-LRT and IRB (research projects)

Institute of Oncology Research

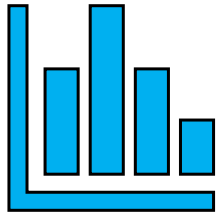
Faculty of Biomedical Sciences

The BCU's current work at IOR



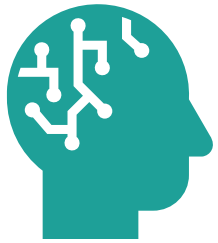
Data Analysis

Bulk and single-cell
RNA-seq
Genomics
Proteomics



Statistics

Experimental design
Power analysis
Survival modelling



Algorithm development & AI

Custom algorithms



Training

Courses and advice for wet-lab and computational researchers

Introduction round



Your name

What do you do? (PhD/ postdoc)

Research species: mouse / human / plants

R expertise: beginner / advanced

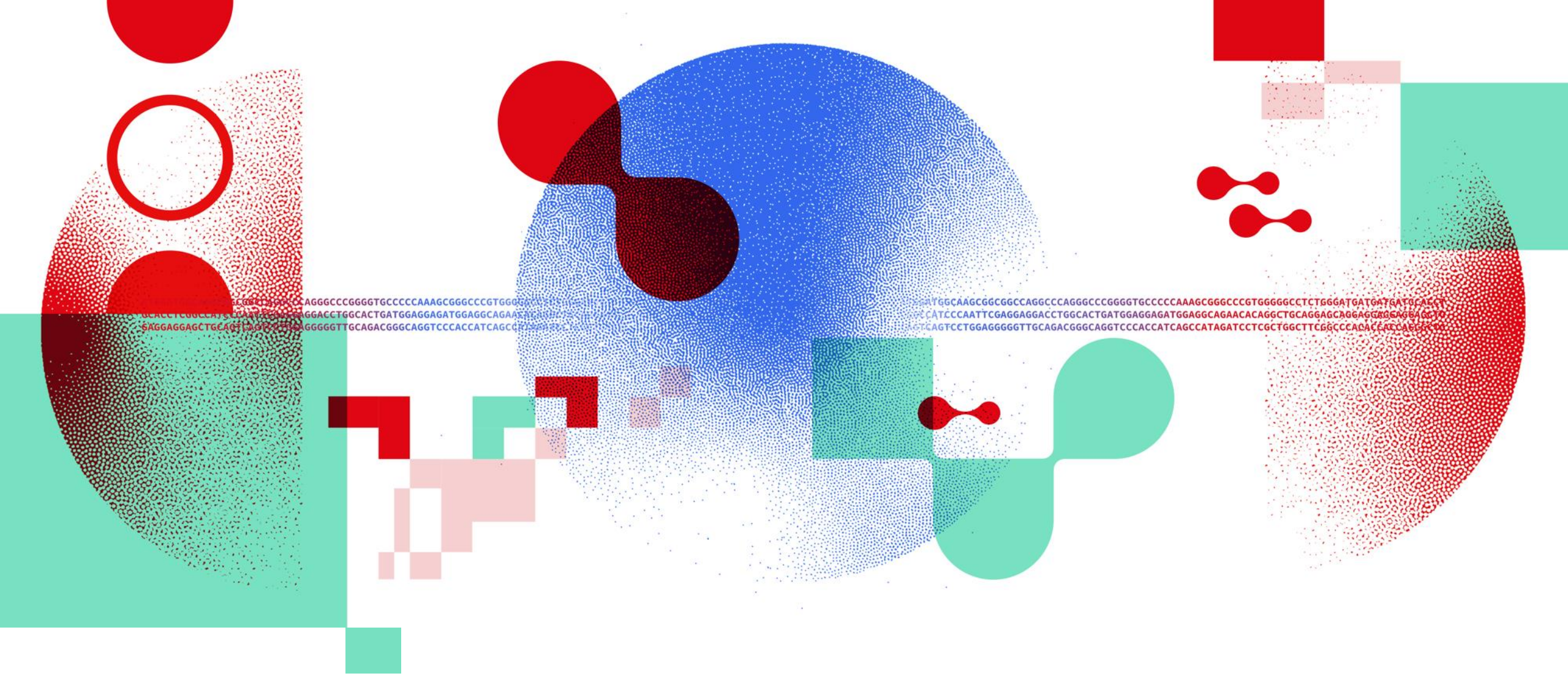
Practical information

Coffee break In the cafeteria, just outside the seminar room.

Aperitif This evening (November 11th), Hotel Unione 6 pm

Lunch Everyone pays for their own lunch.

We can buy food together and eat at the tables at Bios+.



Thank you

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